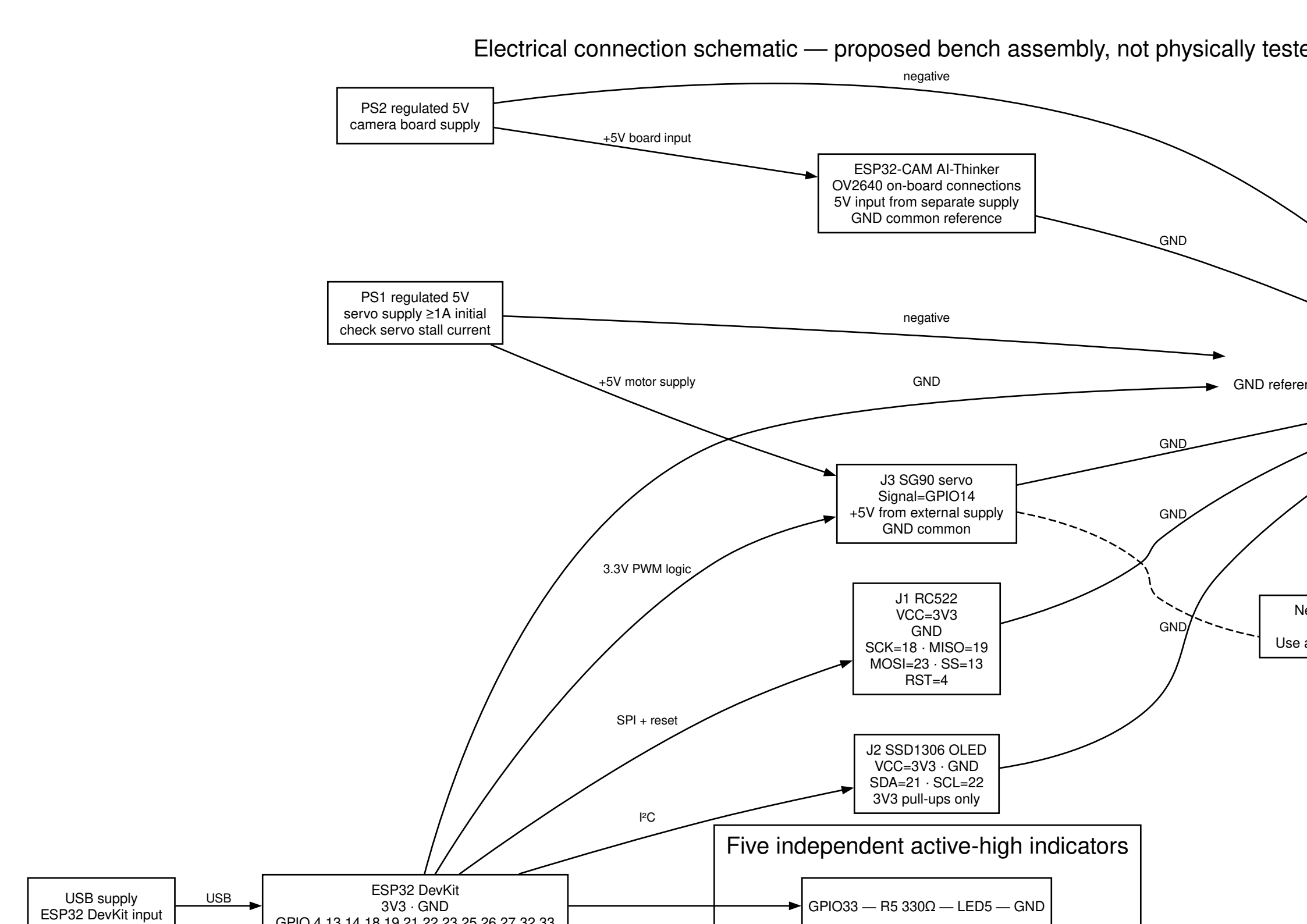
Electrical connection schematic — proposed bench assembly, not physically teste
PS2 regulated 5V
camera board supply
negative
+5V board input
ESP32-CAM AI-Thinker
OV2640 on-board connections
5V input from separate supply
GND common reference
GND
PS1 regulated 5V
servo supply ≥1A initial
check servo stall current
negative
+5V motor supply
GND
GND referen
GND
J3 SG90 servo
Signal=GPIO14
+5V from external supply
GND common
GND
3.3V PWM logic
J1 RC522
VCC=3V3
GND
SCK=18 · MISO=19
MOSI=23 · SS=13
RST=4
GND
Ne
Use a
SPI + reset
J2 SSD1306 OLED
VCC=3V3 · GND
SDA=21 · SCL=22
3V3 pull-ups only
I²C
Five independent active-high indicators
USB supply
ESP32 DevKit input
USB
ESP32 DevKit
3V3 · GND
GPIO 4 13 14 18 19 21 22 23 25 26 27 32 33
GPIO33 — R5 330Ω — LED5 — GND